

write all the possible case when to use long division and partial fraction in z transform

Below is a **complete, exam-oriented classification of all cases** where you must use **long division** and/or **partial-fraction expansion (PFE)** in the **Z-transform**, including **when, why, and in what order**. This is written to remove ambiguity and support correct method selection under time constraints.

1. Fundamental Principle

For a rational Z-transform

$$X(z) = \frac{N(z)}{D(z)}$$

the method you use depends on **two checks**:

1. **Degree check** → decides **long division**
 2. **Factorization check** → decides **partial fractions**
-

2. Long Division — All Possible Cases

Rule (Non-negotiable)

Use **long division** if:

$$\deg(N(z)) \geq \deg(D(z))$$

This must be done **before** partial fractions.

Case L1 — Degree of numerator > degree of denominator

$$\deg N > \deg D$$

Action

- Perform **long division**
- Result:

$$X(z) = \text{Polynomial in } z^{-1} + \frac{\text{Proper rational part}}{D(z)}$$

- Apply PFE only to the remaining proper fraction

Why

- Z-transform inversion tables require **proper rational forms**
-

Case L2 — Degree of numerator = degree of denominator

$$\deg N = \deg D$$

Action

- Perform **long division**
- Result:

$$X(z) = \text{Constant} + \frac{\text{Lower-degree numerator}}{D(z)}$$

Why

- Still an **improper rational function**
 - Constant term corresponds to $\delta[n]$
-

Case L3 — Polynomial only (no denominator after division)

Action

- No PFE needed
- Directly invert using:

$$z^{-k} \leftrightarrow \delta[n - k]$$

3. Partial Fraction Expansion — All Possible Cases

Rule

Use partial fractions if:

$$\deg(N(z)) < \deg(D(z))$$

Case P1 — Distinct real poles

$$D(z) = (1 - a_1 z^{-1})(1 - a_2 z^{-1})$$

Action

$$X(z) = \frac{A}{1 - a_1 z^{-1}} + \frac{B}{1 - a_2 z^{-1}}$$

Why

- Each term directly maps to $a^n u[n]$
-

Case P2 — Repeated poles

$$D(z) = (1 - a z^{-1})^k$$

Action

$$X(z) = \frac{A_1}{1 - a z^{-1}} + \frac{A_2}{(1 - a z^{-1})^2} + \dots$$

Why

- Repeated poles produce $na^n u[n]$ -type terms
-

Case P3 — Complex-conjugate poles

$$D(z) = 1 - 2r \cos \omega z^{-1} + r^2 z^{-2}$$

Action

- Use quadratic PFE or standard cosine/sine inversion forms

Why

- Leads to damped sinusoids in time domain

Case P4 — Left-sided or two-sided sequences (ROC critical)

Action

- Same algebraic PFE
- ROC decides:
 - $u[n]$
 - $u[-n - 1]$
 - two-sided sequence

Why

- Algebra alone is insufficient; ROC determines causality
-

4. Combined Use — Long Division + Partial Fractions

Case C1 — Improper rational + factorizable denominator

Steps (mandatory order)

1. Long division
2. Partial fractions on remainder only

Why

- Skipping step 1 leads to incorrect inversion
-

5. Full Decision Table (Exam-Ready)

Condition	Long Division	Partial Fractions	Order
$\deg N > \deg D$	Yes	Yes (after)	LD $\rightarrow$ PFE
$\deg N = \deg D$	Yes	Yes (after)	LD $\rightarrow$ PFE
$\deg N < \deg D$	No	Yes	PFE only
Polynomial only	Yes	No	LD only

Condition	Long Division	Partial Fractions	Order
Distinct poles	No	Yes	PFE
Repeated poles	No	Yes	PFE
Complex poles	No	Yes	PFE
Left-sided ROC	Depends	Yes	LD → PFE

6. One-Line Memory Rule (Highly Reliable)

If the fraction is improper → long division first.

If the fraction is proper → partial fractions.

ROC decides causality, not algebra.

If you want, I can now:

- Solve **one example of each case**, or
- Create a **flowchart for method selection**, or
- Give **common exam mistakes and how to avoid them**



give example of each case step by step solution

Below are **step-by-step solved examples for every possible case** discussed earlier. Each example is chosen to clearly demonstrate **when** and **why** long division and/or partial-fraction expansion (PFE) is required.

Case L1 — Degree of Numerator > Degree of Denominator

(Long Division → Partial Fractions)

Given

$$X(z) = \frac{2z^2 + 3z + 1}{z + 1}$$

Step 1: Degree check

$$\deg(N) = 2 > \deg(D) = 1 \Rightarrow \text{Long division required}$$

Step 2: Long division

$$\frac{2z^2 + 3z + 1}{z + 1} = 2z + 1$$

Step 3: Inverse Z-transform

$$2z \leftrightarrow 2\delta[n + 1], \quad 1 \leftrightarrow \delta[n]$$

Result

$$x[n] = 2\delta[n + 1] + \delta[n]$$

Case L2 — Degree of Numerator = Degree of Denominator

(Long Division → Partial Fractions)

Given

$$X(z) = \frac{z + 2}{z + 1}$$

Step 1: Degree check

$$\deg(N) = \deg(D) = 1 \Rightarrow \text{Long division}$$

Step 2: Long division

$$\frac{z + 2}{z + 1} = 1 + \frac{1}{z + 1}$$

Step 3: Inverse Z-transform

$$1 \leftrightarrow \delta[n]$$

$$\frac{1}{z+1} = \frac{z^{-1}}{1+z^{-1}} \leftrightarrow (-1)^n u[n-1]$$

Result

$$x[n] = \delta[n] + (-1)^n u[n-1]$$

Case P1 — Proper Rational, Distinct Poles

(Partial Fractions only)

Given

$$X(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})}$$

Step 1: Degree check

Proper fraction → no long division

Step 2: Partial fractions

$$X(z) = \frac{A}{1 - \frac{1}{2}z^{-1}} + \frac{B}{1 - \frac{1}{3}z^{-1}}$$

Solving:

$$A = 3, \quad B = -2$$

Step 3: Inverse Z-transform

$$\frac{1}{1 - az^{-1}} \leftrightarrow a^n u[n]$$

Result

$$x[n] = 3 \left(\frac{1}{2}\right)^n u[n] - 2 \left(\frac{1}{3}\right)^n u[n]$$

Case P2 — Proper Rational, Repeated Poles

Given

$$X(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})^2}$$

Step 1: Partial fraction form

$$X(z) = \frac{A}{1 - \frac{1}{2}z^{-1}} + \frac{B}{(1 - \frac{1}{2}z^{-1})^2}$$

Step 2: Solve coefficients

$$A = 0, \quad B = 1$$

Step 3: Inverse Z-transform

$$\frac{1}{(1 - az^{-1})^2} \leftrightarrow na^n u[n]$$

Result

$$x[n] = n \left(\frac{1}{2}\right)^n u[n]$$

Case P3 — Complex Conjugate Poles

Given

$$X(z) = \frac{1}{1 - 2r \cos \omega z^{-1} + r^2 z^{-2}}$$

Step 1: Recognize standard form

Step 2: Use known inverse

$$\leftrightarrow r^n \frac{\sin((n+1)\omega)}{\sin \omega} u[n]$$

Result

$$x[n] = r^n \frac{\sin((n+1)\omega)}{\sin \omega} u[n]$$

Case C1 — Improper Rational + Factorable Denominator

(Long Division → Partial Fractions)

Given

$$X(z) = \frac{z^2}{z(z-1)}$$

Step 1: Degree check

$$\deg(N) = 2 = \deg(D) \Rightarrow \text{Long division}$$

Step 2: Long division

$$\frac{z^2}{z(z-1)} = 1 + \frac{z}{z(z-1)}$$

Step 3: Partial fractions

$$\frac{z}{z(z-1)} = \frac{1}{z-1}$$

Step 4: Inverse Z-transform

$$\begin{aligned} 1 &\leftrightarrow \delta[n] \\ \frac{1}{z-1} &= \frac{z^{-1}}{1-z^{-1}} \leftrightarrow u[n-1] \end{aligned}$$

Result

$$x[n] = \delta[n] + u[n-1]$$

Case P4 — Same Algebra, Different ROC (Left-Sided)

Given

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}, \quad |z| < \frac{1}{2}$$

Step 1: Identify ROC $\rightarrow$ left-sided

Step 2: Inverse Z-transform

$$x[n] = -\left(\frac{1}{2}\right)^n u[-n-1]$$

Final Summary Table

Case	Method Used	Key Step
Degree $N > D$	Long division $\rightarrow$ PFE	Make proper
Degree $N = D$	Long division $\rightarrow$ PFE	Extract constant
Proper fraction	PFE only	Pole separation
Repeated poles	PFE	Higher-order terms
Complex poles	Standard forms	Sinusoidal
Left-sided	Same algebra	ROC decides

If you want next, I can:

- Provide **exam-style shortcut rules**, or
- Create **one master flowchart**, or
- Give **common mistakes with corrections**



show an exaple where both long division and partial fraction is used should be complex problem for exam

Below is a **single, fully worked, exam-level problem** that **forces you to use both long division and partial-fraction expansion**, includes **multiple poles**, and is representative of what examiners use to test method selection and execution.

Exam-Level Problem

(Long Division and Partial Fractions Required)

Given

$$X(z) = \frac{2z^3 + 3z^2 + z + 1}{z(z-1)(z-2)}, \quad \text{ROC: } |z| > 2$$

Step 1: Degree Check (Why Long Division Is Mandatory)

$$\deg(N) = 3, \quad \deg(D) = 3$$

Since

$$\deg(N) \geq \deg(D),$$

this is an **improper rational Z-transform**.

Conclusion:

👉 Long division must be performed first (no exceptions).

Step 2: Perform Long Division

Divide:

$$\frac{2z^3 + 3z^2 + z + 1}{z^3 - 3z^2 + 2z}$$

Long division result:

$$X(z) = 2 + \frac{9z^2 - 3z + 1}{z(z-1)(z-2)}$$

Step 3: Check the Remainder

$$\deg(9z^2 - 3z + 1) = 2 < 3$$

Now the expression is **proper**.

Conclusion:

👉 Partial-fraction expansion is now allowed.

Step 4: Partial-Fraction Expansion

Assume:

$$\frac{9z^2 - 3z + 1}{z(z-1)(z-2)} = \frac{A}{z} + \frac{B}{z-1} + \frac{C}{z-2}$$

Multiply both sides by $z(z-1)(z-2)$:

$$9z^2 - 3z + 1 = A(z-1)(z-2) + Bz(z-2) + Cz(z-1)$$

Solve for coefficients

Value of z	Result
$z = 0$	$A = \frac{1}{2}$
$z = 1$	$B = -7$
$z = 2$	$C = \frac{25}{2}$

Final Decomposition

$$X(z) = 2 + \frac{1}{2z} - \frac{7}{z-1} + \frac{25}{2(z-2)}$$

Step 5: Convert Each Term to Standard Z-Transform Form

Constant term

$$2 \leftrightarrow 2\delta[n]$$

Term 1

$$\frac{1}{2z} = \frac{1}{2}z^{-1} \leftrightarrow \frac{1}{2}\delta[n-1]$$

Term 2

$$\frac{1}{z-1} = \frac{z^{-1}}{1-z^{-1}} \leftrightarrow u[n-1]$$

Term 3

$$\frac{1}{z-2} = \frac{z^{-1}}{1-2z^{-1}} \leftrightarrow 2^{n-1}u[n-1]$$

Step 6: Final Inverse Z-Transform

$$x[n] = 2\delta[n] + \frac{1}{2}\delta[n-1] - 7u[n-1] + \frac{25}{2}2^{n-1}u[n-1]$$

Why This Is a Perfect Exam Problem

- ✓ Degree equality forces **long division**
 - ✓ Multiple poles force **partial fractions**
 - ✓ ROC confirms **right-sided sequence**
 - ✓ Mix of **impulses + exponentials + steps**
 - ✓ Common exam trap if long division is skipped
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One-Line Examiner Logic

"If the student skips long division, the partial fractions will be wrong."

If you want next, I can:

- Compress this into a **2-minute exam solution**, or
- Give a **variant with repeated poles**, or
- Show how marks are typically allocated for this problem

show this full process no shortcut process

Below is the **complete, no-shortcut, line-by-line solution**, exactly as an examiner expects to see it. Every algebraic step is shown explicitly. Nothing is assumed.

GIVEN (Exam Problem)

$$X(z) = \frac{2z^3 + 3z^2 + z + 1}{z(z-1)(z-2)}, \quad \text{ROC: } |z| > 2$$

STEP 1: Expand the Denominator Fully

You must **never** perform long division without a single polynomial.

$$z(z-1)(z-2)$$

First multiply:

$$(z-1)(z-2) = z^2 - 3z + 2$$

Now multiply by z :

$$z(z^2 - 3z + 2) = z^3 - 3z^2 + 2z$$

So:

$$X(z) = \frac{2z^3 + 3z^2 + z + 1}{z^3 - 3z^2 + 2z}$$

STEP 2: Degree Check (Mandatory Justification)

$$\deg(N) = 3, \quad \deg(D) = 3$$

Since:

$$\deg(N) \geq \deg(D)$$

👉 Long division is compulsory

STEP 3: Long Division (FULL DETAIL)

Divide:

$$\frac{2z^3 + 3z^2 + z + 1}{z^3 - 3z^2 + 2z}$$

First term

$$\frac{2z^3}{z^3} = 2$$

Multiply divisor by 2:

$$2(z^3 - 3z^2 + 2z) = 2z^3 - 6z^2 + 4z$$

Subtract:

$$\begin{aligned} &(2z^3 + 3z^2 + z + 1) \\ &-(2z^3 - 6z^2 + 4z) \\ &= 9z^2 - 3z + 1 \end{aligned}$$

Long division result

$$X(z) = 2 + \frac{9z^2 - 3z + 1}{z^3 - 3z^2 + 2z}$$

STEP 4: Check the Remainder

$$\deg(9z^2 - 3z + 1) = 2 < 3$$

Now the fraction is **proper**.

👉 Partial fractions can now be applied

STEP 5: Factor the Denominator Again

$$z^3 - 3z^2 + 2z = z(z - 1)(z - 2)$$

STEP 6: Write Partial Fraction Assumption (NO SKIPPING)

$$\frac{9z^2 - 3z + 1}{z(z - 1)(z - 2)} = \frac{A}{z} + \frac{B}{z - 1} + \frac{C}{z - 2}$$

STEP 7: Remove Denominator (Critical Step)

Multiply both sides by $z(z - 1)(z - 2)$:

$$9z^2 - 3z + 1 = A(z - 1)(z - 2) + Bz(z - 2) + Cz(z - 1)$$

STEP 8: Expand Every Term Fully

Expand each product

$$(z - 1)(z - 2) = z^2 - 3z + 2$$

$$z(z - 2) = z^2 - 2z$$

$$z(z - 1) = z^2 - z$$

Substitute:

$$9z^2 - 3z + 1 = A(z^2 - 3z + 2) + B(z^2 - 2z) + C(z^2 - z)$$

STEP 9: Collect Like Terms

$$= (A + B + C)z^2 + (-3A - 2B - C)z + 2A$$

STEP 10: Equate Coefficients (NO SHORTCUT)

Compare with:

$$9z^2 - 3z + 1$$

Coefficient equations:

1. z^2 : $A + B + C = 9$
 2. z : $-3A - 2B - C = -3$
 3. Constant: $2A = 1$
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STEP 11: Solve the System Step-by-Step

From equation (3):

$$A = \frac{1}{2}$$

Substitute into (1):

$$\frac{1}{2} + B + C = 9 \Rightarrow B + C = \frac{17}{2}$$

Substitute into (2):

$$-3\left(\frac{1}{2}\right) - 2B - C = -3$$

$$-\frac{3}{2} - 2B - C = -3$$

$$2B + C = \frac{3}{2}$$

Solve the two equations

$$\begin{cases} B + C = \frac{17}{2} \\ 2B + C = \frac{3}{2} \end{cases}$$

Subtract:

$$B = -7$$

Then:

$$C = \frac{17}{2} + 7 = \frac{25}{2}$$

STEP 12: Write the Final Decomposition

$$X(z) = 2 + \frac{1}{2z} - \frac{7}{z-1} + \frac{25}{2(z-2)}$$

STEP 13: Convert Each Term to Inverse Z Form

Term 1

$$2 \leftrightarrow 2\delta[n]$$

Term 2

$$\frac{1}{2z} = \frac{1}{2}z^{-1} \leftrightarrow \frac{1}{2}\delta[n-1]$$

Term 3

$$\frac{1}{z-1} = \frac{z^{-1}}{1-z^{-1}} \leftrightarrow u[n-1]$$

Term 4

$$\frac{1}{z-2} = \frac{z^{-1}}{1-2z^{-1}} \leftrightarrow 2^{n-1}u[n-1]$$

STEP 14: Final Answer (Fully Justified)

$$x[n] = 2\delta[n] + \frac{1}{2}\delta[n-1] - 7u[n-1] + \frac{25}{2}2^{n-1}u[n-1]$$

WHY THIS SOLUTION IS EXAM-PERFECT

- Long division justified
 - Every multiplication expanded
 - No coefficient shortcuts
 - Proper PFE method
 - ROC respected
 - Clean final result
-

If you want next:

- Same problem **with repeated poles**, or
- Same difficulty **but left-sided ROC**, or
- **Marking-scheme version** (how examiners award points)



that means i have to do long division untill the power becomes same

Short answer: No.

That is a **common misconception**.

You **do NOT** do long division “until the powers become the same.”

You do long division **until the remainder becomes a proper fraction**.

Below is the **exact, correct rule**, stated precisely and unambiguously.

Correct Rule (Exam-Safe)

You perform **long division only until:**

$$\deg(\text{remainder numerator}) < \deg(\text{denominator})$$

At that point, you **STOP long division** and then apply **partial fractions**.

What You Should NOT Do

- ✗ Do **not** keep dividing to “match powers”
- ✗ Do **not** force numerator and denominator to have the same degree
- ✗ Do **not** continue once the fraction is already proper

Doing any of the above will **create incorrect terms** and cost marks.

Why This Rule Exists (Conceptual Reason)

- Long division converts an **improper rational function** into:
 - a **polynomial part** (impulses / shifted impulses), plus
 - a **proper rational part** (invertible via tables)
- Partial fractions are defined **only for proper rational functions**

Once the remainder is proper, the job of long division is finished.

Correct Stopping Condition (One Line)

Stop long division as soon as the remainder's degree is lower than the denominator's degree.

Visual Example (Very Clear)

Given

$$\frac{2z^3 + 3z^2 + z + 1}{z^3 - 3z^2 + 2z}$$

After ONE division step:

$$= 2 + \frac{9z^2 - 3z + 1}{z^3 - 3z^2 + 2z}$$

Now:

$$\deg(9z^2 - 3z + 1) = 2 < 3$$

- ✓ Proper fraction achieved
 - ✓ **STOP long division**
 - ✓ Start partial fractions
-

Counterexample (To Kill the Misconception)

Given

$$\frac{z^2 + 1}{z^3 + 2z}$$

Here:

$$\deg(N) = 2 < \deg(D) = 3$$

- ✓ Already proper
- ✓ **NO long division at all**

Even though the “powers are not the same”.

Final Exam Memory Rule (Gold Rule)

Long division $\iff \deg(N) \geq \deg(D)$

Stop when remainder is proper

One-Sentence Answer to Your Question

No—long division is not done until powers match; it is done only until the remainder’s degree is strictly less than the denominator’s degree.

If you want, I can now:

- Give **5 quick true/false exam traps**, or
- Draw a **decision flowchart**, or
- Show **exact marks lost** for this mistake in exams



↳ ### Case L2 — Degree of numerator = degree of denominator $\deg N = \deg D$ $\deg N = \deg D$ **Action** * Perform **long division** * Result: $X(z) = \text{Constant} + \frac{\text{Lower-degree numerator}}{D(z)}$...

show real example of exam question and solve and explain step by step

Below is a **realistic exam-style question** that tests **Case L2: degree of numerator = degree of denominator**, followed by a **fully detailed, step-by-step solution**, with clear explanations of **why each step is required**. This is exactly the level and structure used in Signals & Systems / DSP exams.

Exam Question (Typical University Level)

Question:

Find the inverse Z-transform of the following signal. Show all steps clearly.

$$X(z) = \frac{z + 3}{z - 2}, \quad \text{ROC: } |z| > 2$$

Step 1: Identify the Type of Rational Function (Very Important)

- Numerator: $z + 3 \rightarrow \text{degree} = 1$
- Denominator: $z - 2 \rightarrow \text{degree} = 1$

$$\deg(N) = \deg(D)$$

Conclusion

This is an **improper rational function**.

Therefore, long division is mandatory.

Partial fractions alone are **not allowed** at this stage.

Step 2: Perform Long Division (No Shortcuts)

We divide:

$$\frac{z + 3}{z - 2}$$

First division term

$$\frac{z}{z} = 1$$

Multiply divisor by 1:

$$1 \cdot (z - 2) = z - 2$$

Subtract:

$$(z + 3) - (z - 2) = 5$$

Step 3: Write the Result After Long Division

$$X(z) = 1 + \frac{5}{z - 2}$$

Now check degrees:

- Remainder numerator: degree 0
- Denominator: degree 1

$$\deg(\text{remainder}) < \deg(D)$$

- ✓ Proper fraction achieved
 - ✓ Long division **stops here**
-

Step 4: Interpret Each Term Physically

Constant term

$$1 \leftrightarrow \delta[n]$$

This is why Case L2 always produces a delta function.

Step 5: Prepare the Proper Fraction for Inversion

$$\frac{5}{z-2}$$

Rewrite into standard Z-transform form:

$$\frac{5}{z-2} = 5 \frac{z^{-1}}{1-2z^{-1}}$$

Step 6: Apply the Standard Inverse Z-Transform Pair

Known pair:

$$\frac{z^{-1}}{1-az^{-1}} \leftrightarrow a^{n-1}u[n-1]$$

Here $a = 2$:

$$5 \frac{z^{-1}}{1-2z^{-1}} \leftrightarrow 5 \cdot 2^{n-1}u[n-1]$$

Step 7: Write the Final Answer

$$\boxed{x[n] = \delta[n] + 5 \cdot 2^{n-1}u[n-1]}$$

Why This Is a Perfect Case L2 Example

- Degrees are **equal** → long division required
- Long division produces a **constant** → impulse term
- Remaining fraction is proper → standard inversion
- ROC confirms **right-sided (causal)** signal

Examiner's Logic (What They Expect You to Show)

1. Degree comparison

2. Justification for long division
3. Correct stopping point
4. Proper Z-transform form
5. Correct time-domain interpretation

Missing **Step 2** or **Step 3** typically costs **40–50% of marks**.

One-Line Rule Reinforced

If $\deg N = \deg D$, you must do long division once, and the result will always contain a $\delta[n]$ term.

If you want next, I can:

- Give a **harder Case L2** with quadratic terms, or
- Show a **common wrong solution and why it fails**, or
- Provide a **marking-scheme breakdown** for this question



show an complex one

Below is a **genuinely complex, full-marks exam-level** question that **forces long division + partial fractions + careful inversion**, with **every step explained**. This is harder than typical tutorials and very representative of **Signals & Systems / DSP final exams**.

Complex Exam Question

Question:

Find the inverse Z-transform of

$$X(z) = \frac{z^4 + 2z^3 + 3z^2 + z + 1}{z^2(z-1)(z-2)}, \quad \text{ROC: } |z| > 2$$

Show **all steps** and justify every method used.

STEP 1: Expand the Denominator Fully

You must work with a **single polynomial**.

$$z^2(z - 1)(z - 2)$$

First:

$$(z - 1)(z - 2) = z^2 - 3z + 2$$

Multiply by z^2 :

$$z^2(z^2 - 3z + 2) = z^4 - 3z^3 + 2z^2$$

So,

$$X(z) = \frac{z^4 + 2z^3 + 3z^2 + z + 1}{z^4 - 3z^3 + 2z^2}$$

STEP 2: Degree Check (Why Long Division Is Required)

- Numerator degree = 4
- Denominator degree = 4

$$\deg(N) = \deg(D)$$

Conclusion

This is an **improper rational function**.

$\Rightarrow$ Long division is mandatory

STEP 3: Perform Long Division (NO SKIPPING)

Divide:

$$\frac{z^4 + 2z^3 + 3z^2 + z + 1}{z^4 - 3z^3 + 2z^2}$$

First term

$$\frac{z^4}{z^4} = 1$$

Multiply:

$$1 \cdot (z^4 - 3z^3 + 2z^2) = z^4 - 3z^3 + 2z^2$$

Subtract:

$$\begin{aligned} & (z^4 + 2z^3 + 3z^2 + z + 1) \\ & \quad - (z^4 - 3z^3 + 2z^2) \\ & \quad = 5z^3 + z^2 + z + 1 \end{aligned}$$

Second term

$$\frac{5z^3}{z^4} = \frac{5}{z} \quad (\text{not allowed})$$

So we **stop** here.

Long division result:

$$X(z) = 1 + \frac{5z^3 + z^2 + z + 1}{z^4 - 3z^3 + 2z^2}$$

STEP 4: Simplify the Remainder

Factor out z^2 from the denominator:

$$z^4 - 3z^3 + 2z^2 = z^2(z - 1)(z - 2)$$

So:

$$X(z) = 1 + \frac{5z^3 + z^2 + z + 1}{z^2(z - 1)(z - 2)}$$

Check degrees:

- Remainder numerator degree = 3
- Denominator degree = 4

- ✓ Proper fraction achieved
 - ✓ Long division stops
-

STEP 5: Set Up Partial Fractions (CRITICAL STEP)

Because of repeated pole at $z = 0$:

$$\frac{5z^3 + z^2 + z + 1}{z^2(z-1)(z-2)} = \frac{A}{z} + \frac{B}{z^2} + \frac{C}{z-1} + \frac{D}{z-2}$$

STEP 6: Remove Denominator

Multiply both sides by $z^2(z-1)(z-2)$:

$$5z^3 + z^2 + z + 1 = Az(z-1)(z-2) + B(z-1)(z-2) + Cz^2(z-2) + Dz^2(z-1)$$

STEP 7: Expand Every Term

$$(z-1)(z-2) = z^2 - 3z + 2$$

Expand terms:

- $Az(z^2 - 3z + 2) = A(z^3 - 3z^2 + 2z)$
 - $B(z^2 - 3z + 2)$
 - $C(z^3 - 2z^2)$
 - $D(z^3 - z^2)$
-

STEP 8: Collect Like Terms

$$= (A + C + D)z^3 + (-3A + B - 2C - D)z^2 + (2A - 3B)z + 2B$$

STEP 9: Equate Coefficients

Compare with:

$$5z^3 + z^2 + z + 1$$

System of equations:

1. $A + C + D = 5$
 2. $-3A + B - 2C - D = 1$
 3. $2A - 3B = 1$
 4. $2B = 1$
-

STEP 10: Solve System Step-by-Step

From (4):

$$B = \frac{1}{2}$$

From (3):

$$2A - \frac{3}{2} = 1 \Rightarrow A = \frac{5}{4}$$

Substitute into (1):

$$\frac{5}{4} + C + D = 5 \Rightarrow C + D = \frac{15}{4}$$

Substitute into (2):

$$\begin{aligned} -3\left(\frac{5}{4}\right) + \frac{1}{2} - 2C - D &= 1 \\ -\frac{15}{4} + \frac{1}{2} - 2C - D &= 1 \\ 2C + D &= -\frac{17}{4} \end{aligned}$$

Solve:

$$C = -8, \quad D = \frac{47}{4}$$

STEP 11: Final Decomposition

$$X(z) = 1 + \frac{5}{4z} + \frac{1}{2z^2} - \frac{8}{z-1} + \frac{47}{4(z-2)}$$

STEP 12: Inverse Z-Transform Term by Term

Term	Time Domain
1	$\delta[n]$
$\frac{1}{z}$	$\delta[n - 1]$
$\frac{1}{z^2}$	$\delta[n - 2]$
$\frac{1}{z-1}$	$u[n - 1]$
$\frac{1}{z-2}$	$2^{n-1}u[n - 1]$

STEP 13: Final Answer

$$x[n] = \delta[n] + \frac{5}{4}\delta[n - 1] + \frac{1}{2}\delta[n - 2] - 8u[n - 1] + \frac{47}{4}2^{n-1}u[n - 1]$$

Why This Is Truly “Complex”

- ✓ High-order numerator
 - ✓ Repeated pole
 - ✓ Long division mandatory
 - ✓ Multi-term PFE
 - ✓ Mixed impulses + exponentials
 - ✓ ROC-consistent
-

Final Reinforcement (Key Concept)

You do long division only until the remainder becomes proper — never until “powers match.”

If you want, I can next:

- Show a **left-sided version of this exact problem**, or
- Give a **common wrong solution and explain why it fails**, or
- Convert this into a **10-mark exam answer format**



$$H(z) = \frac{1 + 0.25z^{-1}}{1 + 0.8z^{-1} - 0.84z^{-2}}$$

find out the z-inverse transform explain why long division or partial function is used

We solve this **exactly like an exam question**, and I will **explicitly justify** whether **long division** or **partial fractions** is used.

Given

$$H(z) = \frac{1 + 0.25z^{-1}}{1 + 0.8z^{-1} - 0.84z^{-2}}$$

We are asked to find the **inverse Z-transform** $h[n]$.

Step 1: Decide — Long Division or Partial Fractions?

Write degrees in z^{-1}

- Numerator degree = 1
- Denominator degree = 2

$$\deg(N) < \deg(D)$$

Conclusion (VERY IMPORTANT)

This is already a **proper rational function**.

NO long division is required

👉 Partial-fraction expansion is the correct method

Why:

Long division is used **only when** $\deg(N) \geq \deg(D)$.

Here, that condition is **not satisfied**.

Step 2: Factor the Denominator

Denominator:

$$1 + 0.8z^{-1} - 0.84z^{-2}$$

Assume factorization:

$$(1 - az^{-1})(1 - bz^{-1})$$

Expand:

$$1 - (a + b)z^{-1} + abz^{-2}$$

Match coefficients:

$$\begin{cases} -(a + b) = 0.8 \\ ab = -0.84 \end{cases}$$

Solving:

$$a = 0.6, \quad b = -1.4$$

So:

$$1 + 0.8z^{-1} - 0.84z^{-2} = (1 - 0.6z^{-1})(1 + 1.4z^{-1})$$

Step 3: Set Up Partial Fractions (NO SHORTCUT)

$$H(z) = \frac{1 + 0.25z^{-1}}{(1 - 0.6z^{-1})(1 + 1.4z^{-1})}$$

Assume:

$$H(z) = \frac{A}{1 - 0.6z^{-1}} + \frac{B}{1 + 1.4z^{-1}}$$

Step 4: Remove Denominator

Multiply both sides by

$$(1 - 0.6z^{-1})(1 + 1.4z^{-1}):$$

$$1 + 0.25z^{-1} = A(1 + 1.4z^{-1}) + B(1 - 0.6z^{-1})$$

Step 5: Expand and Collect Terms

$$= A + B + (1.4A - 0.6B)z^{-1}$$

Match coefficients with:

$$1 + 0.25z^{-1}$$

Coefficient equations:

1. $A + B = 1$
 2. $1.4A - 0.6B = 0.25$
-

Step 6: Solve the System

From (1):

$$B = 1 - A$$

Substitute into (2):

$$1.4A - 0.6(1 - A) = 0.25$$

$$1.4A - 0.6 + 0.6A = 0.25$$

$$2A = 0.85 \Rightarrow A = 0.425$$

$$B = 1 - 0.425 = 0.575$$

Step 7: Final Z-Domain Decomposition

$$H(z) = \frac{0.425}{1 - 0.6z^{-1}} + \frac{0.575}{1 + 1.4z^{-1}}$$

Step 8: Inverse Z-Transform (Using Standard Pairs)

Known pair:

$$\frac{1}{1 - az^{-1}} \leftrightarrow a^n u[n]$$

Term 1

$$\frac{0.425}{1 - 0.6z^{-1}} \leftrightarrow 0.425(0.6)^n u[n]$$

Term 2

$$\frac{0.575}{1 + 1.4z^{-1}} = \frac{0.575}{1 - (-1.4)z^{-1}} \leftrightarrow 0.575(-1.4)^n u[n]$$

 **Final Answer**

$$\boxed{h[n] = 0.425(0.6)^n u[n] + 0.575(-1.4)^n u[n]}$$

Final Conceptual Summary (Exam Gold)

Why NO long division?

- Degree of numerator < degree of denominator
- Already a proper rational function

Why partial fractions?

- Denominator factorizes into first-order terms
 - Required for table-based inverse Z-transform
-

One-Line Rule Reinforced

If the Z-transform is proper → use partial fractions.
If it is improper → use long division first.

If you want, I can:

- Redo this assuming a **left-sided ROC**, or
- Show a **common wrong solution and why it fails**, or
- Convert this into a **10-mark exam answer format**

